

WAD Paper Solution

Q1 (a) Short Questions (1 mark × 3 = 3 Marks)

1. Define Microsoft Intermediate Language (MSIL).

MSIL is the intermediate set of CPU-independent instructions generated by .NET compilers. It contains metadata and instructions for object creation, memory handling, and method calls. At runtime, CLR converts MSIL into native machine code using JIT compiler.

2. What is the use of the global.asax file?

The global.asax file defines application-level and session-level events. Example events:

- Application_Start → executes when application starts.
- Session_Start → executes when a new user session begins.
It helps in managing global settings like authentication, logging, and error handling.

3. Define ViewState in ASP.NET.

ViewState is a mechanism for preserving control values between postbacks in ASP.NET. It stores data in a hidden field on the page encoded in Base64. This allows controls like textboxes to maintain values even after reloading the page.

Q1 (b) Objective / MCQs / T-F / Fill in blanks (1 mark × 7 = 7 Marks)

1. CLR converts MSIL code into → **Native Code** ✓
2. ASP.NET is a client-side scripting technology → **False** ✓
3. Validator ensures field is not empty → **RequiredFieldValidator** ✓
4. Assemblies in .NET can be **Private or Shared** ✓
5. global.asax is used for → **Application-level events** ✓
6. State management technique storing data in browser → **ViewState** ✓
7. CLS stands for → **Common Language Specification** ✓

Q2 (a) Two Questions of 2 Marks = 4 Marks

1. Differentiate between Managed and Unmanaged code. (2 Marks)

- **Managed Code**
 - Runs under CLR.
 - Provides garbage collection, type safety, exception handling.
 - Example: C# code compiled in .NET.
- **Unmanaged Code**
 - Runs outside CLR.
 - Developer must handle memory allocation and security.
 - Example: C/C++ programs compiled to machine code.

2. Short note on ASP.NET. (2 Marks)

ASP.NET is a Microsoft web framework for building dynamic websites and web applications.

- Built on the .NET platform.
 - Supports server-side controls, event-driven programming, and MVC pattern.
 - Provides features like authentication, state management, caching, and deployment support.
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Q2 (b) Two Questions of 3 Marks = 6 Marks

1. Importance & process of Garbage Collection in .NET (3 Marks)

- **Importance:** Frees unused memory, prevents memory leaks, improves performance.
- **Process:**
 1. **Marking** → identifies live objects in memory.
 2. **Relocation/Sweep** → removes unused objects.
 3. **Compaction** → reorganizes memory to improve efficiency.
- CLR automatically manages memory using Garbage Collector.

2. What are Cookies in ASP.NET? (3 Marks)

Cookies are small data files stored on client browser used to store user information.

- **Types:**

1. **Persistent Cookies** → stored on disk with expiry date.
 2. **Non-Persistent (Session) Cookies** → deleted after session ends.
- **Uses:** Maintain login sessions, user preferences, shopping cart data.
 - **Example:**

```
Response.Cookies["username"].Value = "John";
```

```
Response.Cookies["username"].Expires = DateTime.Now.AddDays(1);
```

Q3 Attempt any TWO (5 Marks each = 10 Marks)

1. ASP.NET Validation Controls (5 Marks)

Types with examples:

- **RequiredFieldValidator** → ensures input is not empty. Example: validating username.
- **RangeValidator** → checks input is within range (e.g., age between 18–60).
- **CompareValidator** → compares two values (e.g., password and confirm password).
- **RegularExpressionValidator** → validates format (e.g., email: abc@gmail.com).
- **CustomValidator** → user-defined validation using server-side code.

2. State Management Techniques (5 Marks)

- **Client-side:**
 - **ViewState** → stores data in hidden field on page.
 - **Cookies** → stores data in browser.
 - **QueryString** → passes data via URL.
 - **Hidden Fields** → invisible storage in form.
- **Server-side:**
 - **Session State** → stores data per user session.
 - **Application State** → stores global data for all users.
 - **Cache** → temporary data for performance.

3. Difference between CTS & CLS (5 Marks)

- **CTS (Common Type System)**
 - Defines all types supported by CLR.
 - Ensures data type compatibility across languages.
 - Example: defines int32, float, string.
 - **CLS (Common Language Specification)**
 - Subset of CTS.
 - Defines rules to ensure cross-language interoperability.
 - Example: avoids unsigned types not supported in all languages.
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Q4 (5 Marks each = 10 Marks)

(a) Just-In-Time (JIT) Compiler in .NET (5 Marks)

- JIT compiler converts MSIL code into native machine code at runtime.
- **Types of JIT:**
 1. **Pre-JIT** → compiles all code at once at deployment.
 2. **Econo-JIT** → compiles only methods called, discards after use (saves memory).
 3. **Normal JIT** → compiles on demand and keeps cached code for reuse.
- **Advantage:** Improves performance, enables platform independence.

(b) Client-side vs Server-side State Management (5 Marks)

- **Client-side** (data stored in browser):
 - ViewState, Cookies, QueryString, Hidden Fields.
 - Light load on server, but less secure.
- **Server-side** (data stored on server):
 - Session State, Application State, Cache, Database.
 - More secure, suitable for sensitive data, but consumes server resources.

OR

(b) Master Pages & Nested Master Pages (5 Marks)

- **Master Page:** Defines common layout (header, footer, menu). Content pages plug into placeholders.
- **Nested Master Pages:** A Master Page can itself be based on another Master Page → supports hierarchical layout.

Example:

- Main Master Page: defines header, footer.
- Nested Master Page: adds side menu inside main layout.
- Content Page: uses placeholders of nested master.